

Booth Multiplication

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Multiplication - Unsigned

1011	Multiplicand (11)
×1101	Multiplier (13)
1011	}
0000	
1011	
1011	}
10001111	
	Product (143)

- Using paper – pencil method

► Inferences

- Multiplication involves the generation of partial products, one for each digit in the multiplier. These partial products are then summed to produce the final product.
- The partial products are easily defined. When the multiplier bit is 0, the partial product is 0. When the multiplier is 1, the partial product is the multiplicand.
- The total product is produced by summing the partial products. For this operation, each successive partial product is shifted one position to the left relative to the preceding partial product.
- The multiplication of two n-bit binary integers results in a product of up to 2n bits in length.

Comparison of Multiplication of Unsigned and Twos Complement Integers

- ▶ We have seen that addition and subtraction can be performed on numbers in twos complement notation by treating them as unsigned integers.
- ▶ Unfortunately, this simple scheme will not work for multiplication.
- ▶ Ex: Multiply 11 (1011) by 13 (1101) to get 143 (10001111).
- ▶ If we interpret these as twos complement numbers, we have -5 (1011) times -3 (1101) equals -15 (10001111).
- ▶ This example demonstrates that straightforward multiplication will not work if both the multiplicand and multiplier are negative.
- ▶ In fact, it will not work if either the multiplicand or the multiplier is negative

Comparison of Multiplication of Unsigned and Twos Complement Integers

$\begin{array}{r} 1001 \quad (9) \\ \times 0011 \quad (3) \\ \hline 00001001 \quad 1001 \times 2^0 \\ 00010010 \quad 1001 \times 2^1 \\ \hline 00011011 \quad (27) \end{array}$	$\begin{array}{r} 1001 \quad (-7) \\ \times 0011 \quad (3) \\ \hline 11111001 \quad (-7) \times 2^0 = (-7) \\ 11110010 \quad (-7) \times 2^1 = (-14) \\ \hline 11101011 \quad (-21) \end{array}$
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Unsigned integers

Twos complement integers

Serial addition

20 00010100
30 × 00011110
00000000
00010100
00010100
00010100
00010100
00000000
00000000
00000000
000001001011000
600

Booth algorithm

20 00010100
30 × 00011110
111111111101100
00000010100
0000001001011000
600

Booth's Multiplication Algorithm

$$\begin{aligned}M \times (00011110) &= M \times (2^4 + 2^3 + 2^2 + 2^1) \\&= M \times (16 + 8 + 4 + 2) \\&= M \times 30\end{aligned}$$

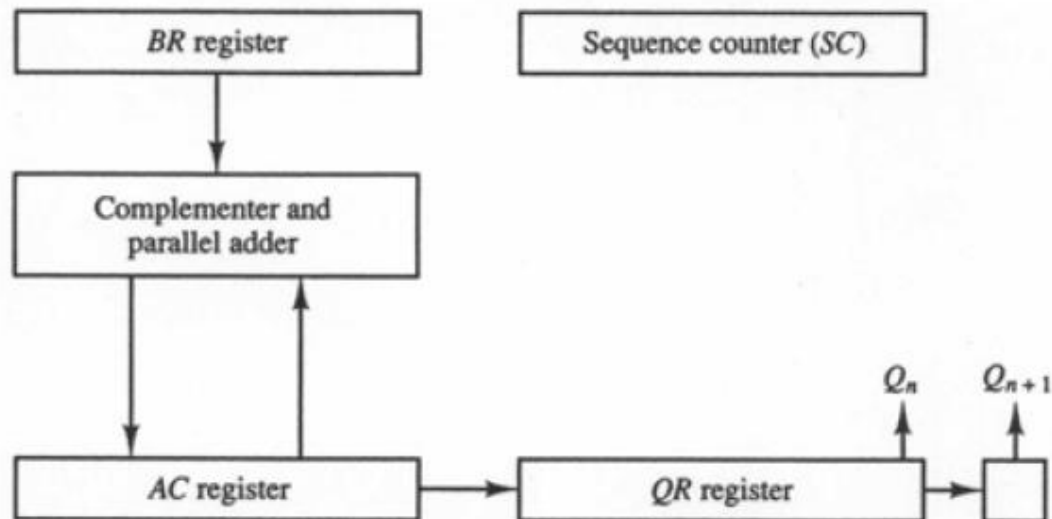
The number of operations can be reduced to two if we observe that:

$$2^n + 2^{n-1} + \dots + 2^{n-k} = 2^{n+1} - 2^{n-k}$$

$$\begin{aligned}M \times (00011110) &= M \times (2^5 - 2^1) \\&= M \times (32 - 2) \\&= M \times 30\end{aligned}$$

$$\begin{aligned}14 = 01110 &= 2^4 - 2^1 \\&= 16 - 2\end{aligned}$$

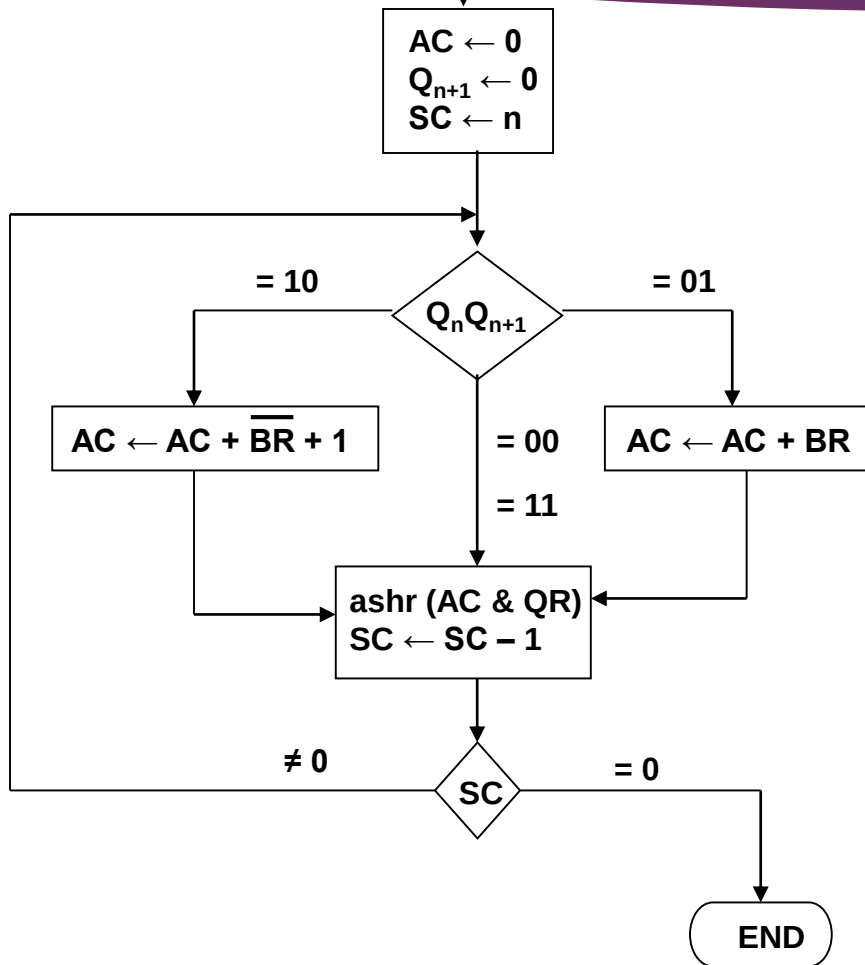
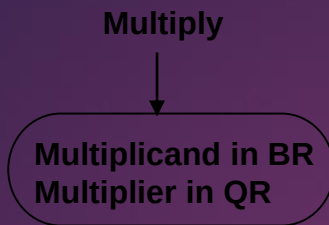
Hardware for Booth's Multiplication



Flowchart for Booth Multiplication

Example: $-9 \times -13 = 117$

BR = 10111, $\overline{BR} + 1 = 01001$



Comment	AC	QR	Q _{n+1}	SC
	00000	10011	0	5
Subtract BR	01001			
	01001			
Ashr	00100	11001	1	4
Ashr	00010	01100	1	3
Add BR	10111			
	11001			
Ashr	11100	10110	0	2
Ashr	11110	01011	0	1
Subtract BR	01001			
	00111			
Ashr	00011	10101	1	0

Booth Multiplier Recording Table

Multiplier		Version of Multiplicand selected by bit i
Bit i	Bit i + 1	
0	0	$0 \times M$
0	1	$+1 \times M$
1	0	$-1 \times M$
1	1	$0 \times M$

Exercise

- Show the step by step multiplication process using Booth algorithm when the following binary numbers are multiplied. Assume 5-bit registers that hold signed numbers. The multiplicand in both cases is +15.

$$(+15) \times (+13)$$

$$(+15) \times (-13)$$

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- ▶ 00011110 (30)
 - ▶ $Q_n=0Q_{n+1}=0=0$
 - ▶ $10 = -1$
 - ▶ $11=0$
 - ▶ $11=0$
 - ▶ $11=0$
 - ▶ $01=1$
 - ▶ $00=0$
 - ▶ $00=0$
 - ▶ 001000-10(30)